

ECE 532/COS 572/MAE 572/ROB 532
Safety-Critical Robotics and AI
Problem Set <X>

<Your name>

September 15, 2026

Problem 1

Citations and math. In the following, we recall that *all models are wrong, but some are useful* [1].

1. The discrete-time system dynamics are $x_{t+1} = f_t(x_t, u_t, d_t)$.
2. The inevitable backward-reachable tube of the failure set is $\overleftarrow{\mathcal{R}}_{t:T}^-(\mathcal{F}) \subset \mathbb{R}^n$.

Including code. You can use `\lstinputlisting` to include code snippets/files if you like. Check out https://www.overleaf.com/learn/latex/Code_listing for details. The code below is rendered directly from the example file `steer.py` included with this template.

```
1 def steer(pos_goal, pos_now, d_max):
2     """Moves towards pos_goal up to maximum distance d_max."""
3
4     # Compute distance between pos_now and pos_goal.
5     distance = ... # *YOUR CODE HERE*
6     if distance > d_max:
7         # Compute x_new.
8         x_new = ... # *YOUR CODE HERE*
9     else:
10         x_new = pos_goal
11     return x_new
```

From a notebook to your write-up. L^AT_EX won't read `.ipynb` files directly, but you can instead save each function you want to show into its own `.py` file, drop those files to your L^AT_EX project folder, and include them exactly as above. This is easy with a file-export cell at the end of your Jupyter/Colab notebook:

```
1 import inspect
2 from google.colab import files # Colab only; drop outside Colab.
3
4 SUBMIT = ["steer"] # Functions to show in your write-up.
5
6 for name in SUBMIT:
7     fn = globals().get(name)
8     if fn is not None:
9         with open(f"{name}.py", "w") as f:
10             f.write(inspect.getsource(fn))
11     files.download(f"{name}.py") # Colab only; drop outside Colab.
```

This writes your implementation of `steer` to `steer.py`, ready to sit next to `main.tex` and be included as above. Please include only the functions a problem actually asks for, rather than a full notebook export.

References

- [1] G. E. P. Box, "Science and Statistics," *Journal of the American Statistical Association*, vol. 71, no. 356, pp. 791–799, 1976.